

$$c = a + jb = |c| e^{j\phi}$$

$$|c| = \sqrt{a^2 + b^2}$$

$$\phi = \tan^{-1}\left(\frac{b}{a}\right)$$

(phase always in radians for EM waves)

1)  $c_1 = 2 + j$ ,  $c_2 = -2 + j3$

(a)  $c_1 + c_2$

$$\begin{aligned} c_1 + c_2 &= (2 + j) + (-2 + j3) \\ &= (2 + (-2)) + j(1 + 3) \\ &= 0 + j4 \end{aligned}$$

$$c = 0 + j4$$

$$|c| = \sqrt{0^2 + 4^2} = \sqrt{16} = 4$$

$$\phi = \tan^{-1}\left(\frac{4}{0}\right) = \frac{\pi}{2}$$

Part (a)

ANSWER

$$c_1 + c_2 = 0 + j4 = 4e^{j\frac{\pi}{2}} = 4 \angle 90^\circ$$

(b)  $c_1 - c_2$

$$\begin{aligned} c_1 - c_2 &= (2 + j) - (-2 + j3) \\ &= (2 - (-2)) + j(1 - 3) \\ &= 4 - j2 \end{aligned}$$

$$c = 4 - j2$$

$$|c| = \sqrt{4^2 + (-2)^2} = \sqrt{16 + 4} = \sqrt{20}$$

$$|c| = 2\sqrt{5}$$

$$\phi = \tan^{-1}\left(\frac{-2}{4}\right) = -0.464$$

Part (b)

Answer

$$c_1 - c_2 = 4 - j2 = 2\sqrt{5} e^{-j0.464} = 2\sqrt{5} \angle -26.6^\circ$$

(c)  $c_1 c_2$

$$c_1 = 2 + j$$

$$c_2 = -2 + j3$$

$$|c_1| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$|c_2| = \sqrt{(-2)^2 + (3)^2} = \sqrt{13}$$

$$\phi_1 = \tan^{-1}\left(\frac{1}{2}\right) = 0.464$$

$$\phi_2 = \tan^{-1}\left(\frac{3}{-2}\right) = 2.159$$

$$c_1 = \sqrt{5} e^{j0.464}$$

$$c_2 = \sqrt{13} e^{j2.159}$$

$$c_1 c_2 = |c_1| |c_2| e^{j(\phi_1 + \phi_2)}$$

$$= (\sqrt{5})(\sqrt{13}) e^{j(0.464 + 2.159)}$$

$$= \sqrt{65} e^{j2.623}$$

$$c = \sqrt{65} e^{j2.623}$$

$$a = |c| \cos(\phi)$$

$$b = |c| \sin(\phi)$$

$$a = \sqrt{65} \cos(2.623) = -7$$

$$b = \sqrt{65} \sin(2.623) = 4$$

Part (c)

Answer

$$c_1 c_2 = -7 + j4 = \sqrt{65} e^{j2.623} = \sqrt{65} \angle 150.3^\circ$$

(d)  $\frac{c_1}{c_2}$

From part (c),  $c_1 = \sqrt{5} e^{j0.464}$   $c_2 = \sqrt{13} e^{j2.159}$

$$\frac{c_1}{c_2} = \frac{|c_1|}{|c_2|} e^{j(\theta_1 - \theta_2)}$$

$$= \frac{\sqrt{5}}{\sqrt{13}} e^{j(0.464 - 2.159)}$$

$$= \sqrt{\frac{5}{13}} e^{-j1.695}$$

$$c = \sqrt{\frac{5}{13}} e^{-j1.695}$$

$$a = |c| \cos(\theta)$$

$$b = |c| \sin(\theta)$$

$$a = \left(\sqrt{\frac{5}{13}}\right) \cos(-1.695) = -0.077$$

$$b = \left(\sqrt{\frac{5}{13}}\right) \sin(-1.695) = -0.615$$

$$\frac{c_1}{c_2} = -0.077 - j0.615 = \sqrt{\frac{5}{13}} e^{-j1.695} = \sqrt{\frac{5}{13}} \angle -97.1^\circ$$

Part (d)  
Answer

2) (a)  $z^2 = 1 + j$   $|c| = \sqrt{(1)^2 + (1)^2} = \sqrt{2}$

$$c = 1 + j$$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$$

$$z^2 = c$$

$$z = c^{1/2} = (|c| e^{j\theta})^{1/2} = (|c| e^{j(\theta + 2\pi n)})^{1/2} = \sqrt{|c|} e^{j(\frac{\theta}{2} + n\pi)} = (\sqrt{|c|} e^{j\theta/2}) e^{jn\pi} = \pm \sqrt{c}$$

$$n = 0$$

$$z = (\sqrt{|c|} e^{j\theta/2}) e^{j(0)\pi}$$

$$z = \sqrt{|c|} e^{j\theta/2}$$

$$z = \sqrt{\sqrt{2}} e^{j\frac{\pi}{4}}$$

$$z = 2^{1/4} e^{j\frac{\pi}{8}}$$

$$z_1 = 1.189 e^{j\frac{\pi}{8}}, z_2 = 1.189 e^{j\frac{9\pi}{8}}$$

$$n = 1$$

$$z = (\sqrt{|c|} e^{j\theta/2}) e^{j(1)\pi}$$

$$z = \sqrt{|c|} e^{j(\theta/2 + \pi)}$$

$$z = \sqrt{\sqrt{2}} e^{j(\frac{\pi}{2} + \pi)}$$

$$z = 2^{1/4} e^{j\frac{9\pi}{8}}$$

$$a_1 = |z_1| \cos\left(\frac{\pi}{8}\right) = 1.098 \quad a_2 = |z_2| \cos\left(\frac{9\pi}{8}\right) = -1.098$$

$$b_1 = |z_1| \sin\left(\frac{\pi}{8}\right) = 0.455 \quad b_2 = |z_2| \sin\left(\frac{9\pi}{8}\right) = -0.455$$

$$Z = 1.098 + j0.455$$

$$Z = -1.098 - j0.455$$

Part (a)

Answer

$$(b) (j4)^{1/2} = (0+j4)^{1/2}$$

$$|c| = \sqrt{0^2 + 4^2} = 4$$

$$\theta = \tan^{-1}\left(\frac{4}{0}\right) = \frac{\pi}{2} \text{ or } 90^\circ$$

$$(4e^{j\frac{\pi}{2}})^{1/2} = 2e^{j\frac{\pi}{4}} \text{ or } 2e^{j\frac{5\pi}{4}}$$

$\downarrow$                        $\downarrow$   
 $2\angle 45^\circ$  ,     $2\angle 225^\circ$

$$z_1 = 2\left(\cos\left(\frac{\pi}{4}\right) + j\sin\left(\frac{\pi}{4}\right)\right) \quad z_2 = 2\left(\cos\left(\frac{5\pi}{4}\right) + j\sin\left(\frac{5\pi}{4}\right)\right)$$

$$z_1 = \sqrt{2} + j\sqrt{2}$$

$$z_2 = -\sqrt{2} - j\sqrt{2}$$

$$\boxed{z_1 = \sqrt{2} + j\sqrt{2}}$$

$$\boxed{z_2 = -\sqrt{2} - j\sqrt{2}}$$

Part (b)  
Answer

$$(c) (1+j2)^3$$

$$\begin{aligned} (1+j2)(1+j2)(1+j2) &= (1+j4-4)(1+j2) \\ &= (-3+j4)(1+j2) \\ &= -3-j2-8 \\ &= -11-j2 \end{aligned}$$

$$\boxed{(1+j2)^3 = -11-j2}$$

Part (c)  
Answer

$$3) (a) F(t) = 3\sin(\omega t - \frac{\pi}{2})$$

$$F(t) = 3\cos(\omega t - \frac{\pi}{2} - \frac{\pi}{2})$$

$$F(t) = 3\cos(\omega t - \pi)$$

$$\theta = -\pi, F_0 = 3$$

$$F = F_0(\cos(-\pi) + j\sin(-\pi))$$

$$F = -3$$

$$\boxed{F = -3 + j0}$$

Part (a)



$$(b) G(t) = -2\cos(\omega t)$$

$$\theta = 0, G_0 = -2$$

$$G = G_0(\cos(0) + j\sin(0))$$

$$G = -2$$

$$\boxed{G = -2 + j0}$$

Part (b)



$$(c) H(t) = 4 \cos(60\pi t - \frac{\pi}{4})$$

$$\phi = -\frac{\pi}{4}, H_0 = 4$$

$$H = H_0 (\cos(-\frac{\pi}{4}) + j \sin(-\frac{\pi}{4}))$$

$$H = 2\sqrt{2} - j2\sqrt{2}$$

$$H = 2\sqrt{2} - j2\sqrt{2} \quad \swarrow \text{Part (c)}$$

$$(d) E(t) = 4 \sin(\omega t) + \cos(\omega t)$$

$$f(t) = 4 \sin(\omega t)$$

$$= 4 \cos(\omega t - \frac{\pi}{2})$$

$$g(t) = \cos(\omega t)$$

$$\phi_2 = 0, g_0 = 1$$

$$E = 1 - j4 \quad \swarrow \text{Part (d)}$$

$$\phi_1 = -\frac{\pi}{2}, f_0 = 4$$

$$f = 4 (\cos(-\frac{\pi}{2}) + j \sin(-\frac{\pi}{2}))$$

$$g = 1 (\cos(0) + j \sin(0))$$

$$f = -j4$$

$$g = 1$$

$$E = f + g = 1 - j4$$

$$4) (a) K = 1 - j$$

$$|K| = \sqrt{(1)^2 + (-1)^2} = \sqrt{2}$$

$$\phi = \tan^{-1}(\frac{-1}{1}) = -\frac{\pi}{4}$$

$$K(t) = \sqrt{2} \cos(\omega t - \frac{\pi}{4})$$

$$K(t) = \sqrt{2} \cos(\omega t - \frac{\pi}{4}) \quad \swarrow \text{Part (a)}$$

$$(b) K = e^{j0.4}$$

$$\phi = 0.4, |K| = 1$$

$$K(t) = \cos(\omega t + 0.4)$$

$$K(t) = \cos(\omega t + 0.4) \quad \swarrow \text{Part (b)}$$

$$(c) K = 2e^{-j\frac{\pi}{4}} + 3e^{j0.3}$$

$$\phi_1 = -\frac{\pi}{4}, \phi_2 = 0.3$$

$$|K_1| = 2, |K_2| = 3$$

$$K(t) = 2 \cos(\omega t - \frac{\pi}{4}) + 3 \cos(\omega t + 0.3)$$

$$K(t) = 2 \cos(\omega t - \frac{\pi}{4}) + 3 \cos(\omega t + 0.3) \quad \swarrow \text{Part (c)}$$



$$5) \underline{A} = \underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}, \underline{B} = -\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}$$

↙ Part (a)

(a)  $\underline{A} + \underline{B}$

$$\begin{aligned} \underline{A} + \underline{B} &= (\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) + (-\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) \\ &= (1-1)\underline{\hat{x}} + (-2-2)\underline{\hat{y}} + (3+3)\underline{\hat{z}} \\ &= 0\underline{\hat{x}} - 4\underline{\hat{y}} + 6\underline{\hat{z}} \end{aligned}$$

$$\underline{A} + \underline{B} = 0\underline{\hat{x}} - 4\underline{\hat{y}} + 6\underline{\hat{z}}$$

(b)  $\underline{A} - \underline{B}$

$$\begin{aligned} \underline{A} - \underline{B} &= (\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) - (-\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) \\ &= (1-(-1))\underline{\hat{x}} + (-2-(-2))\underline{\hat{y}} + (3-3)\underline{\hat{z}} \\ &= 2\underline{\hat{x}} + 0\underline{\hat{y}} + 0\underline{\hat{z}} \end{aligned}$$

$$\underline{A} - \underline{B} = 2\underline{\hat{x}} + 0\underline{\hat{y}} + 0\underline{\hat{z}}$$

(c)  $\underline{A} \cdot \underline{B}$

$$\begin{aligned} \underline{A} \cdot \underline{B} &= (\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) \cdot (-\underline{\hat{x}} - 2\underline{\hat{y}} + 3\underline{\hat{z}}) \\ &= (1)(-1) + (-2)(-2) + (3)(3) \\ &= -1 + 4 + 9 = 12 \end{aligned}$$

$$\underline{A} \cdot \underline{B} = 12$$

(d)  $\underline{A} \times \underline{B}$

$$\begin{array}{c} \underline{A} \\ \underline{B} \end{array} \begin{vmatrix} \underline{\hat{x}} & \underline{\hat{y}} & \underline{\hat{z}} \\ 1 & -2 & 3 \\ -1 & -2 & 3 \end{vmatrix} = (-2(3) - 3(-2))\underline{\hat{x}} - (1(3) - 3(-1))\underline{\hat{y}} + (1(-2) - (-2)(-1))\underline{\hat{z}}$$

$$= 0\underline{\hat{x}} - 6\underline{\hat{y}} - 4\underline{\hat{z}}$$

$$\underline{A} \times \underline{B} = 0\underline{\hat{x}} - 6\underline{\hat{y}} - 4\underline{\hat{z}}$$

↙ Part (d)

$$6) \quad \underline{A}(t) = \cos(\omega t) \underline{\hat{x}} - 2 \cos(\omega t - \frac{\pi}{2}) \underline{\hat{y}} - 2 \sin(\omega t) \underline{\hat{z}} \\ = \cos(\omega t) \underline{\hat{x}} - 2 \cos(\omega t - \frac{\pi}{2}) \underline{\hat{y}} - 2 \cos(\omega t - \frac{\pi}{2}) \underline{\hat{z}}$$

$$A_x = 1, A_y = -2, A_z = -2$$

$$\phi_x = 0, \phi_y = -\frac{\pi}{2}, \phi_z = -\frac{\pi}{2}$$

$$\underline{A} = e^{j0} \underline{\hat{x}} - 2e^{j\frac{\pi}{2}} \underline{\hat{y}} - 2e^{j\frac{\pi}{2}} \underline{\hat{z}}$$

Part (a)

$$\underline{A} = e^{j0} \underline{\hat{x}} - 2e^{-j\frac{\pi}{2}} \underline{\hat{y}} - 2e^{-j\frac{\pi}{2}} \underline{\hat{z}}$$

$$(b) \quad \underline{A}(t) = (\sin(\omega t) - 3 \sin(\omega t)) \underline{\hat{y}} + (\sin(\omega t) - \cos(\omega t)) \underline{\hat{z}} \\ = (-2 \sin(\omega t)) \underline{\hat{y}} + (\cos(\omega t - \frac{\pi}{2}) - \cos(\omega t)) \underline{\hat{z}} \\ = -2 \cos(\omega t - \frac{\pi}{2}) \underline{\hat{y}} + (\cos(\omega t - \frac{\pi}{2}) - \cos(\omega t)) \underline{\hat{z}}$$

$$\cos(\omega t - \frac{\pi}{2}) - \cos(\omega t) = -j - 1 \\ = -1 - j$$

$$|c| = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$\phi = \tan^{-1}(\frac{-1}{-1}) = -\frac{3\pi}{4}$$

$$A_y = -2, A_z = \sqrt{2}$$

$$\phi_y = -\frac{\pi}{2}, \phi = -\frac{3\pi}{4}$$

Part (b)

$$\underline{A} = -2e^{-j\pi/2} \underline{\hat{y}} + \sqrt{2}e^{-j3\pi/4} \underline{\hat{z}}$$

$$(c) \quad \underline{A}(t) = 0.3 \sin(ky - \omega t) \underline{\hat{x}} \\ = 0.3 \sin(-\omega t + ky) \underline{\hat{x}} \\ = 0.3 \cos(-\omega t + ky - \frac{\pi}{2}) \underline{\hat{x}} \\ = 0.3 \cos(-( \omega t - ky + \frac{\pi}{2} )) \underline{\hat{x}} \\ = 0.3 \cos(\omega t - ky + \frac{\pi}{2}) \underline{\hat{x}}$$

$$A_x = 0.3, \phi_x = -ky + \frac{\pi}{2}$$

Part (c)

$$\underline{A} = 0.3 e^{j(-ky + \frac{\pi}{2})} \underline{\hat{x}}$$

$$7) \underline{K} = j\hat{x} + \hat{z} = (0+j)\hat{x} + (1+j0)\hat{z}$$

(a)

$$K_x = \sqrt{0^2 + 1^2} = 1, K_z = \sqrt{1^2 + 0^2} = 1$$

$$\phi_x = \tan^{-1}\left(\frac{1}{0}\right) = \frac{\pi}{2}, \phi_z = \tan^{-1}\left(\frac{0}{1}\right) = 0 \quad \swarrow \text{Part (a)}$$

$$\underline{K}(t) = \cos(\omega t + \frac{\pi}{2})\hat{x} + \cos(\omega t)\hat{z}$$

$$(b) \underline{K} = j(\hat{x} + j\hat{z}) = j\hat{x} - \hat{z} = (0+j)\hat{x} + (-1+j0)\hat{z}$$

$$K_x = \sqrt{0^2 + 1^2} = 1, K_z = \sqrt{(-1)^2 + 0^2} = 1$$

$$\phi_x = \tan^{-1}\left(\frac{1}{0}\right) = \frac{\pi}{2}, \phi_z = \tan^{-1}\left(\frac{0}{-1}\right) = \pi$$

$\swarrow$  Part (b)

$$\underline{K}(t) = \cos(\omega t + \frac{\pi}{2})\hat{x} + \cos(\omega t + \pi)\hat{z}$$

$$(c) \underline{K} = e^{-jk_y}\hat{x} - je^{-jk_y}\hat{z}$$

$$\underline{K}(t) = \text{Re}\{\underline{K} e^{j\omega t}\}$$

$$= \text{Re}\{(e^{-jk_y}\hat{x} - je^{-jk_y}\hat{z})e^{j\omega t}\}$$

$$= \text{Re}\{e^{j(\omega t - k_y)}\hat{x} - je^{j(\omega t - k_y)}\hat{z}\}$$

$$= \text{Re}\{e^{j(\omega t - k_y)}\hat{x} + e^{j(\omega t - k_y - \frac{\pi}{2})}\hat{z}\}$$

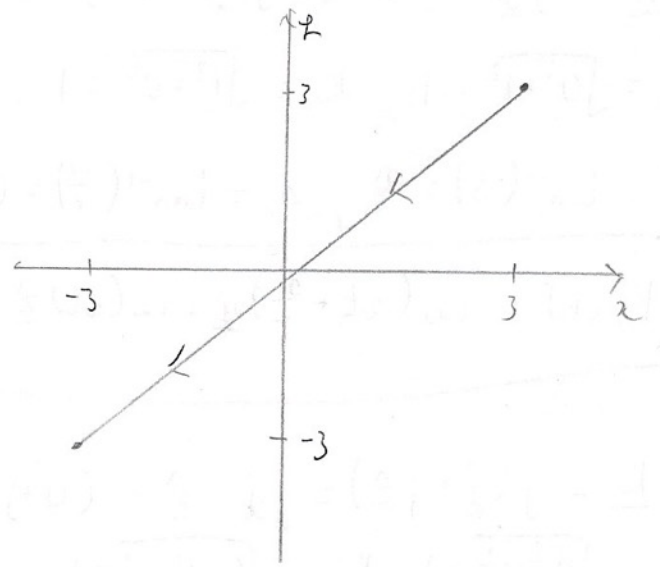
$\swarrow$  Part (c)

$$\underline{K}(t) = \cos(\omega t - k_y)\hat{x} + \sin(\omega t - k_y)\hat{z}$$

$$8) (a) \underline{A} = 3\hat{x} + 3\hat{y} = 3e^{j0}\hat{x} + 3e^{j0}\hat{y}$$

$$\begin{aligned}\underline{A}(t) &= \text{Re}\{\underline{A} e^{j\omega t}\} \\ &= \text{Re}\{(3e^{j0}\hat{x} + 3e^{j0}\hat{y}) e^{j\omega t}\} \\ &= \text{Re}\{3e^{j\omega t}\hat{x} + 3e^{j\omega t}\hat{y}\}\end{aligned}$$

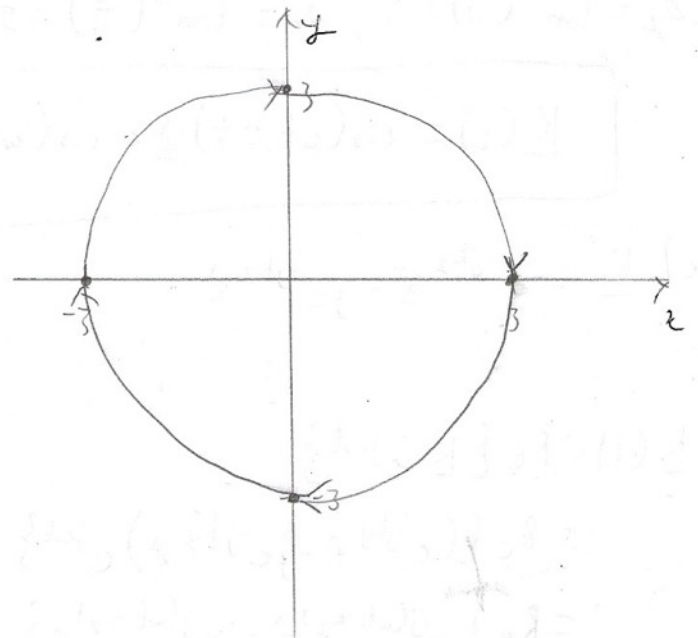
$$\underline{A}(t) = 3\cos(\omega t)\hat{x} + 3\cos(\omega t)\hat{y}$$



$$(b) \underline{A} = 3\hat{x} + j3\hat{y} = 3e^{j0}\hat{x} + 3e^{j\frac{\pi}{2}}\hat{y}$$

$$\begin{aligned}\underline{A}(t) &= \text{Re}\{\underline{A} e^{j\omega t}\} \\ &= \text{Re}\{(3e^{j0}\hat{x} + 3e^{j\frac{\pi}{2}}\hat{y}) e^{j\omega t}\} \\ &= \text{Re}\{3e^{j\omega t}\hat{x} + 3e^{j(\omega t + \frac{\pi}{2})}\hat{y}\}\end{aligned}$$

$$\underline{A}(t) = 3\cos(\omega t)\hat{x} - 3\sin(\omega t)\hat{y}$$



$$(c) \underline{A} = 2\hat{x} + j3\hat{y} = 2e^{j0}\hat{x} + 3e^{j\frac{\pi}{2}}\hat{y}$$

$$\underline{A}(t) = 2\cos(\omega t)\hat{x} - 3\sin(\omega t)\hat{y}$$

